sports ev

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Thoughts

This document has my thoughts and the math used in this project. This project first started off as a group project from Boiler Quant. However, due to my slight sports betting addiction, I wanted to do some more research and improvement. This project estimates a fair probability from a trusted book, in this case FanDuel and bets against other books that are mispriced relative to that estimate.

American Sportsbooks quote odds in "American" format, e.g -115 or +120. The odds will be converted into decimal odds as they are much easier to work with algebraically:

$$d = \begin{cases} 1 + \dfrac{100}{|a|} & \text{if } a < 0 \\ 1 + \dfrac{a}{100} & \text{if } a > 0 \end{cases}$$

a is the American odds, and d being the decimal odds. 1 bet at decimal odds d returns \$$d$ total if it wins (your original dollar plus $d-1$ profit), and \$0 if it loses. For example, FanDuel lists Dillon Brooks Over 28.5 PRA at -111.

$$d = 1 + \frac{100}{111} = 1.901$$

A \$100 bet on this returns \$190.10 if it wins.

1 Problem

Decimal odds convert directly to an implied probability:

$$p_{\text{implied}} = \frac{1}{d}.$$

For the example above, $p_{\text{implied}} = 1/1.901 = 0.526$, or 52.6%. However, sportsbooks don't offer fair odds (obviously). They build in a profit margin called the vig (short for vigorish) by pricing both sides of a bet so their implied probabilities add up to more than 100%. Looking at Dillon Brooks' full line: Over is -111 ($p = 0.526$) and Under is -115 ($p = 0.535$). These sum to 1.061, not 1.0. That extra 6.1% is the book's edge, split across both sides. This means we can't just use p_{implied} directly as our estimate of the true probability because we first need to remove the vig.

1.1 Step 3: Removing the vig

We treat the sharp book as our source of truth, and remove its vig using the proportional method, dividing each side's implied probability by the total, so they sum to exactly 1.

$$p_{\text{over}} = \frac{p_{\text{over, implied}}}{p_{\text{over, implied}} + p_{\text{under, implied}}}, \qquad p_{\text{under}} = \frac{p_{\text{under, implied}}}{p_{\text{over, implied}} + p_{\text{under, implied}}}$$

This is the simplest of several de-vigging methods, but it is reasonable. It assumes the book's edge is distributed proportionally across both sides rather than concentrated on one. So now that we have a fair probability p from the sharp book, we check every the other book's price on the same bet. If another book offers decimal odds d for the same side, the expected profit per dollar staked is:

$$\text{EV} = p \cdot (d - 1) - (1 - p) \cdot 1$$

With probability p we win $(d - 1)$ profit, and with probability $(1 - p)$ we lose our \$1 stake. If EV > 0, this bet has positive expected value – the other book is offering a better price than what we believe is fair. Suppose BetMGM offers Dillon Brooks Over 28.5 PRA at +105 ($d = 2.05$), better than FanDuel's implied price. Using our fair probability $p = 0.496$:

$$\text{EV} = 0.496 \times (2.05 - 1) - 0.504 \times 1 = 0.496 \times 1.05 - 0.504 = 0.521 - 0.504 = 0.017$$

This is a 1.7% expected edge per dollar staked, so a +EV opportunity by our model.

2 Kelly Criterion

Once we've found a positive-EV bet, we need to decide how much to actually bet. Using decimal odds d, define $b = d - 1$. The Kelly fraction of bankroll to bet is derived by taking the derivative, setting it to zero:

$$f^* = \frac{pb - (1 - p)}{b}$$

So continuing from the previous example $p = 0.496$, $b = 1.05$:

$$f^* = \frac{0.496 \times 1.05 - 0.504}{1.05} = \frac{0.521 - 0.504}{1.05} = \frac{0.017}{1.05} = 0.016$$

Full Kelly recommends staking 1.6% of bankroll. In the actual implementation we use fractional Kelly (a quarter of f^*) and cap the maximum stake at a fixed percent of bankroll regardless of edge size, since f^* assumes our fair-probability estimate is exactly correct, and de-vigging is itself an approximation.

3 How does it work?

Scrape live player prop odds from multiple books (FanDuel, BetMGM, ProphetX), all in American odds format. Convert every price to decimal odds, then implied probability. For the sharp book, de-vig each two-sided market (Over/Under) to get a fair probability estimate. For every other book offering the same bet, compute EV using that fair probability. Filter to bets above a minimum EV threshold, then size each with fractional Kelly, capped at a max bankroll percentage.

4 Sources

Clarke, S., Kovalchik, S., & Ingram, M. (2017). "Adjusting Bookmaker's Odds to Allow for Overround." *American Journal of Sports Science*, 5(6), 45–49. – direct source for the proportional de-vig method implemented in Step 3.

Shin, H.S. (1992). "Prices of State Contingent Claims with Insider Traders, and the Favourite-Longshot Bias." *Economic Journal*, 102(411), 426–435. – paper talking about de-vig. do note that Shin's method is way more sophisticated than the proportional method used here.

Strumbelj, E. (2014). "On Determining Probability Forecasts from Betting Odds." – why proportional method is used.

Thorp, E.O. (1969). "Optimal Gambling Systems for Favorable Games." *Review of the International Statistical Institute*, 37(3), 273–293. – applies the Kelly criterion directly to real betting systems, closer to this project's use case than Kelly's original paper.

Kelly, J.L. (1956). "A New Interpretation of Information Rate." Bell System Technical Journal, 35(4), 917–926. – original derivation of the Kelly fraction f^* used in Step 5.